Yisheng Jiang

Full-stack agentic AI engineer

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SUMMARY

Full-stack engineer who ships AI camera, music and video products end to end, from architecture to the App Store. Builds native iOS apps in SwiftUI with on-device ML (Core ML, ONNX Runtime), browser video and audio on WebAssembly (ffmpeg.wasm, Rust, SIMD, AudioWorklet), and the Node/TypeScript backends, nginx and telemetry behind them. Local first, with the cloud as a deliberate fallback. Publishing open-source code on GitHub since 2012.

SELECTED PROJECTS

FinalCap - AI video editor in the browser (grepawk.com | github.com/yishengjiang99/finalcut)

- Chat-driven editor: plain-English requests are mapped to 45 FFmpeg editing tools that run in the browser with ffmpeg.wasm, so video files are not uploaded to edit.
- In-browser captions (on-device Whisper model), translation and lyric captions; Lighthouse performance 99 mobile / 100 desktop. iOS version (SwiftUI) in development.

ffmpeg.wasm fork - multi-threaded FFmpeg 8 core (github.com/yishengjiang99/ffmpeg.wasm)

- Fixed a worker-pool deadlock that hung jobs on machines with 8+ cores, an x264 call-signature crash that had capped the encoder at 4 threads, and an audio-filter uninitialized read that failed repeated jobs.
- Measured on one Apple M4 Pro: re-encode of 10 s of 1080p went from 9.8 s (1 thread) to 1.6 s (14 threads). Memory now starts at 256 MB and grows on demand to 4 GB. (write-up)

filmcraft fork - WebAssembly SIMD (Rust) (github.com/yishengjiang99/filmcraft)

- Added wasm SIMD and a 4-pixel YUV to RGBA kernel plus an adaptive per-frame render budget. Benchmarked 1080p frame conversion about 11-18% faster than upstream (x86 VM and Apple M4 Pro) with bit-identical output. (write-up)

ProTune AI Camera (iOS, live on the App Store) (App Store | photo.grepawk.com | GitHub)

- SwiftUI camera app: on-device Vision reads the scene, an LLM picks a recipe, and the app writes real shutter, ISO, EV, white balance and focus to the camera. Node/TypeScript backend, subscriptions (ProTune Pro, 1.4), push notifications.

AI Music Radar and AI Camera - Music Reader (iOS) (music.grepawk.com | earsheet | omr-sheet-cam)

- Music Radar: fine-tuned Basic Pitch model in Core ML transcribes what you play or sing into engraved sheet music on the iPhone; export MIDI, MusicXML, PDF.
- Music Reader: photo of printed sheet music to MIDI playback with highlighting, using ONNX Runtime (Core ML provider) on-device, no upload. Shared Swift SoundFont engine: sf2player-swift.

gbk (SF2 + MIDI explorer) and xheatgrid (Chrome extension)

- gbk: SoundFont synth and MIDI player in the browser, running in an AudioWorklet off the main thread; exports songs to WAV. xheatgrid: Chrome MV3 extension in plain JS (no build step, no runtime dependencies) that draws a GitHub-style heatmap of daily Posts and Replies on X analytics.

SKILLS

Languages: Swift, TypeScript, JavaScript, Rust, C, Python

Mobile / ML: SwiftUI, AVFoundation, Vision, Core ML, ONNX Runtime, on-device Whisper, LLM tool calling (Grok)

Web / media: WebAssembly (threads, SIMD, SharedArrayBuffer), ffmpeg.wasm, FFmpeg, Web Audio / AudioWorklet, Chrome extensions (MV3), Playwright / Chrome CDP benchmarking

Backend / ops: Node.js (Express), nginx, Linux server operations, telemetry and analytics, CI/CD with GitHub Actions to TestFlight and App Store Connect

EXPERIENCE

GrepAwk LLC, Palo Alto, CA

- Design, build, ship and operate the products above: web app and iOS apps, backend services and server infrastructure.

AppDog (appdog.com) - Sole Engineer 2011–2013

- Only engineer on the platform; kept it running and reliable.